ASSIGNMENT-19.1

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BATCH : 12

Task 1 – Python to Java Conversion

Task:

Translate a simple Python class into Java using AI assistance.

Instructions:

• Prompt AI to convert a Python class with attributes and methods into equivalent Java code.

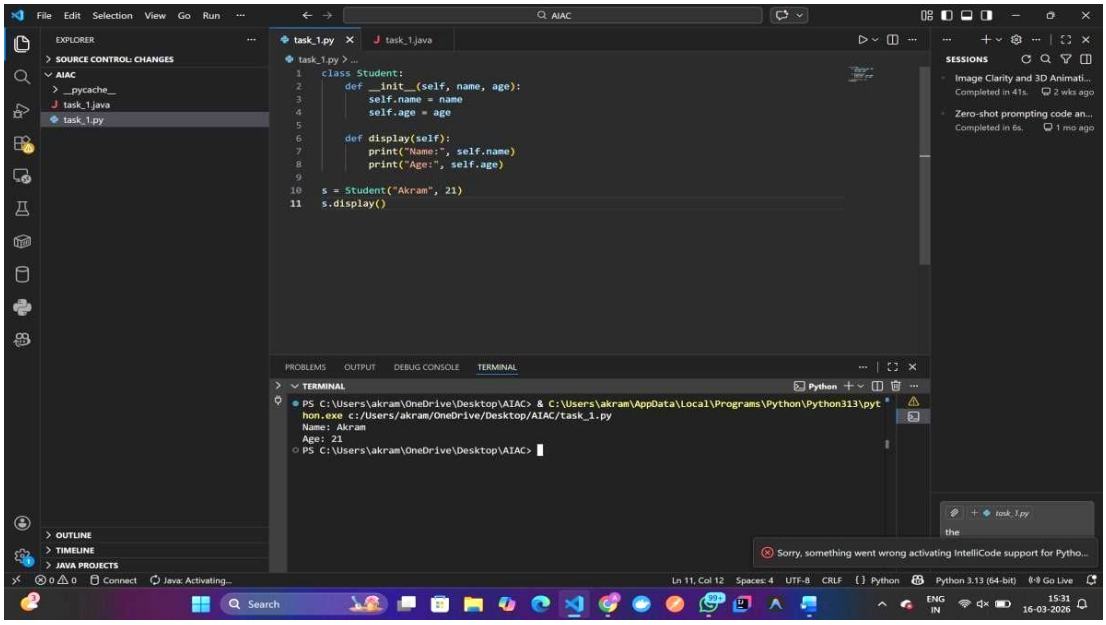
• Verify that constructors, method signatures, and data types are properly handled.

• Run both versions (Python & Java) to compare expected outputs.

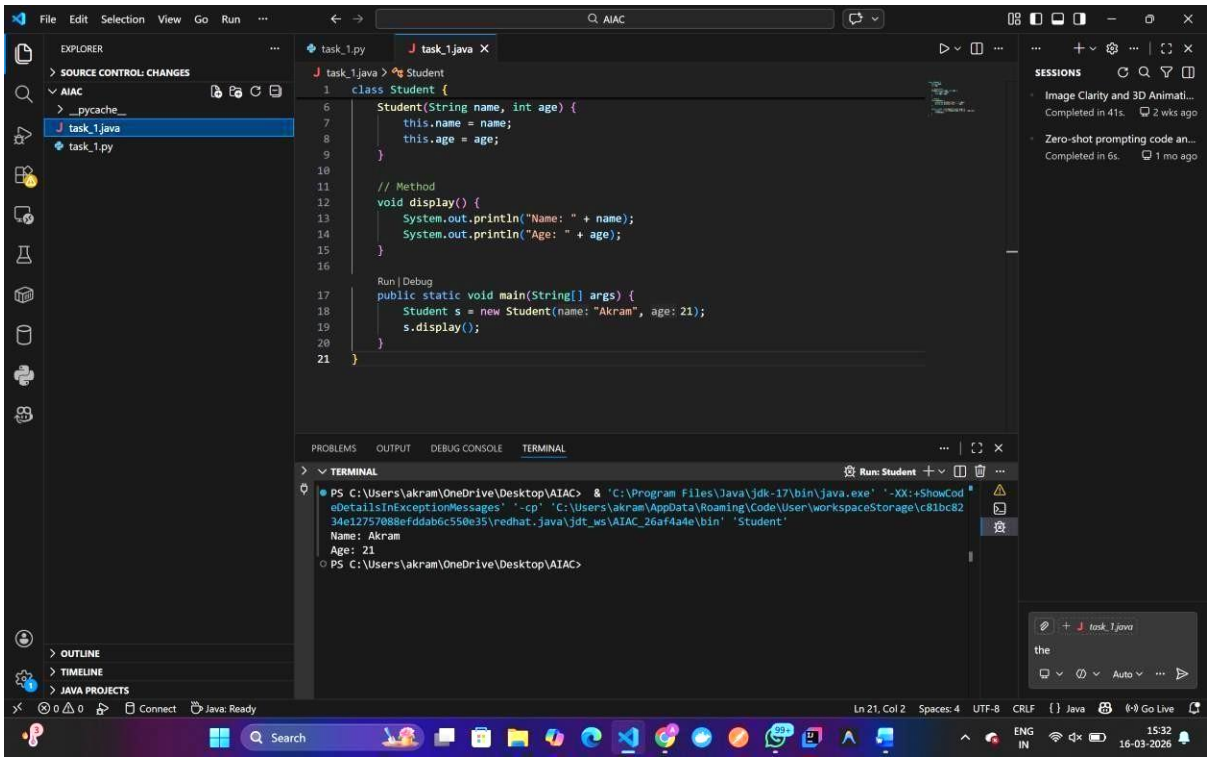
Expected Output:

• Equivalent working Java class translated from Python.

CODE



OUTPUT



EXPLANATION

In this task, a simple Python class was translated into Java using AI assistance. The goal was to understand how object-oriented concepts like attributes, constructors, and methods differ between the two languages. After translation, both the Python and Java programs were executed to ensure they produced the same output. This activity helped in recognizing syntactic differences while maintaining the same logic and functionality. It also demonstrated how AI tools can speed up cross-language development.

Task 2 – C++ to Python Function Conversion

Task:

Convert a C++ function for factorial calculation into Python.

Instructions:

• Ask AI to translate a recursive C++ factorial

function into Python.

• Validate correctness by testing with sample inputs.

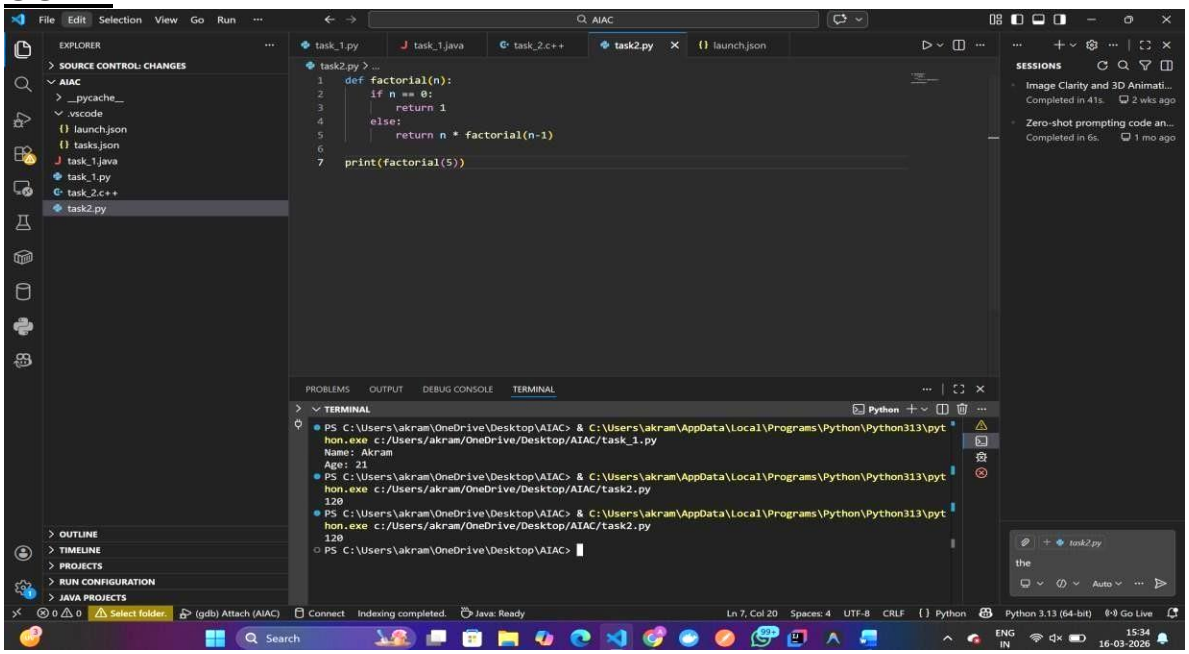
• Ensure Python version follows Pythonic

conventions. Expected Output:

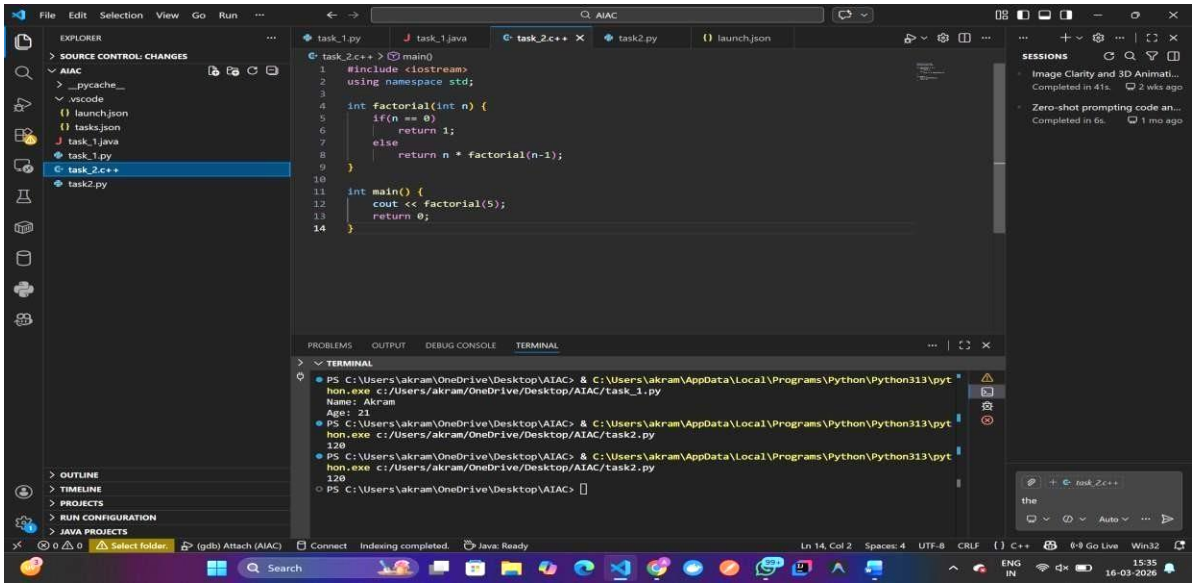
• A correct Python implementation equivalent to the

original C++ factorial function.

CODE



OUTPUT



EXPLANATION

This task involved converting a recursive factorial function written in C++ into Python. The translated Python code was tested with sample inputs to verify that it produced the same results as the original C++ version. Special attention was given to Pythonic conventions such as indentation and the absence of explicit data type declarations. Through this exercise, the differences in syntax and simplicity between C++ and Python became clear. The task highlighted how AI can assist in accurately translating algorithms across

Task 3 – SQL to Pandas Query

Task:

Translate a **SQL query** into a **Pandas DataFrame operation** in Python.

Instructions:

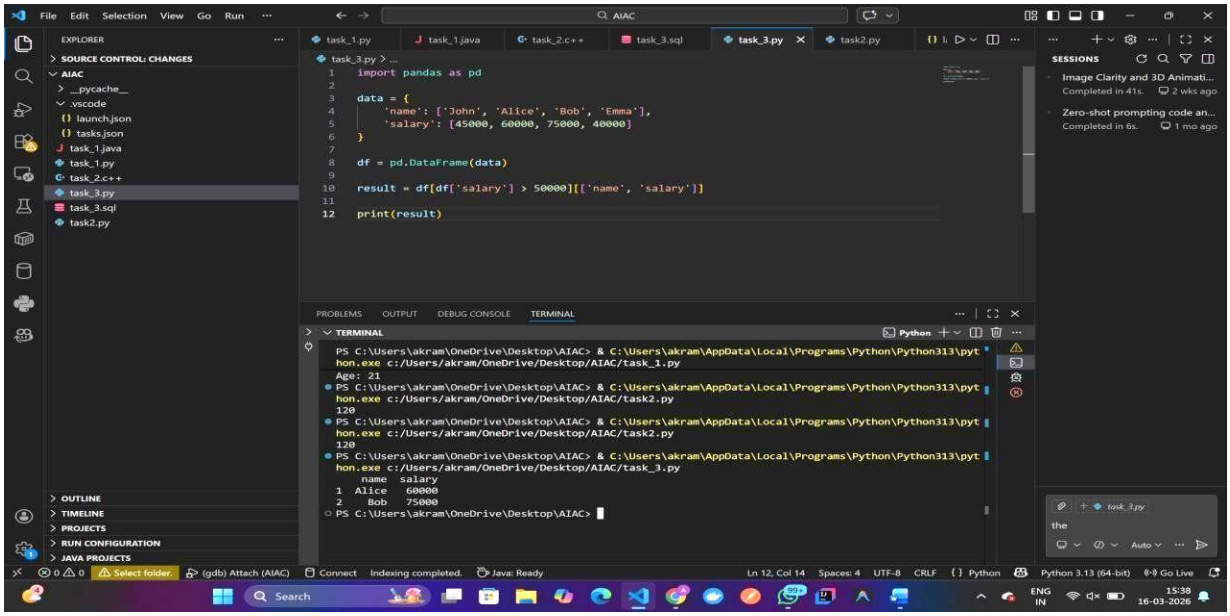
- Provide AI with a SQL query (e.g., SELECT name, salary FROM employees WHERE salary > 50000).
- Ask AI to generate equivalent Pandas code.
- Test the generated code on a sample DataFrame.

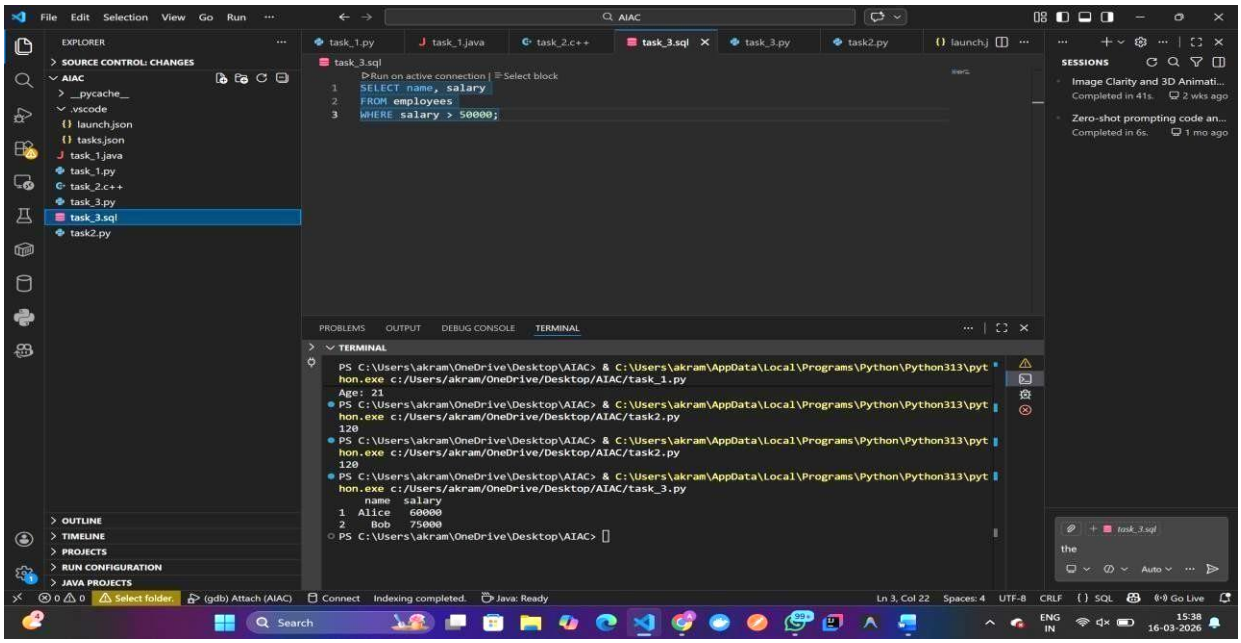
Expected Output:

- Equivalent Pandas query that retrieves the correct

subset of data.

CODE:





EXPLANATION

In this task, a SQL query used to filter employee data was translated into an equivalent Pandas DataFrame operation in Python. The objective was to understand how database-style queries can be implemented using Python for data analysis. A sample DataFrame was created, and the generated Pandas code was executed to ensure it returned the correct subset of data. This exercise demonstrated the similarities between SQL and Pandas while emphasizing their syntactic differences. It also showed how AI can bridge the gap between database and programming environments.

Task 4 – Real-Time Application: Multi-Language Snippet Converter

Scenario:
Students will choose a real-time use case where they need code in multiple languages.
Example:

- Converting a Python web scraping snippet into JavaScript for browser automation.
- Translating a Java method into C# for enterprise applications.

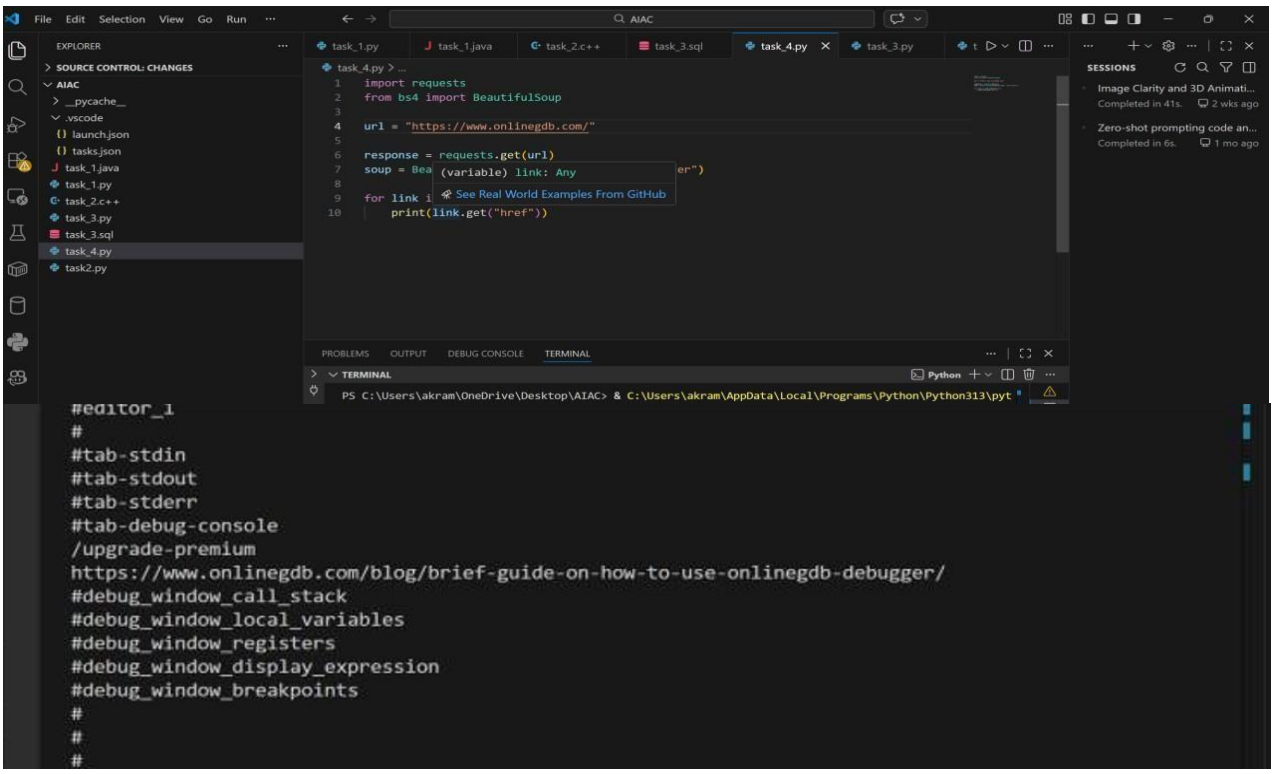
Instructions:

- Prompt AI to translate code between two chosen languages.
- Test both implementations to ensure they produce the same results.
- Document challenges faced during translation (e.g., syntax, libraries).

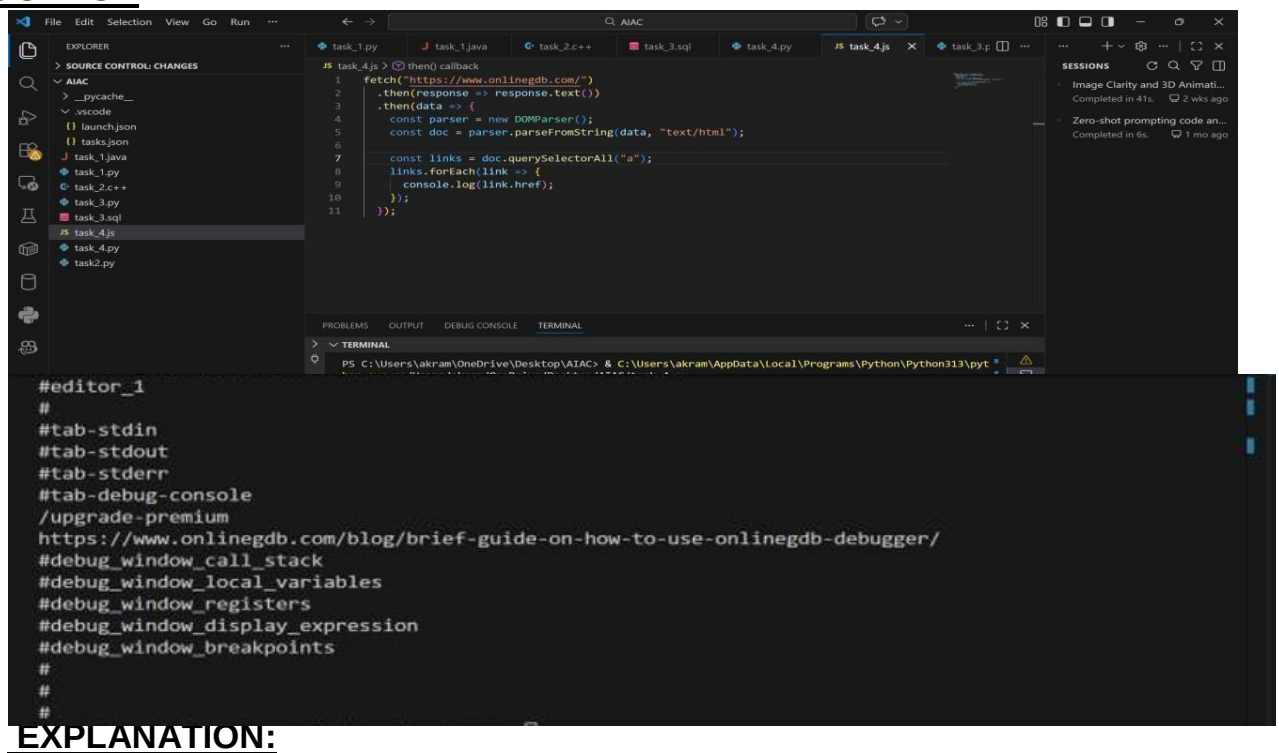
Expected Output:

- Working equivalent code snippets in at least two different programming languages.

CODE:



OUTPUT



EXPLANATION:

This task focused on a real-world scenario where code needs to be available in multiple programming languages. A practical example, such as converting a Python web scraping script into JavaScript, was implemented using AI assistance. Both versions of the code were tested to ensure they produced the same results. The exercise highlighted challenges such as differences in libraries, syntax, and runtime environments. Overall, the task demonstrated how AI can facilitate efficient code translation for cross-platform and enterprise applications.